COURSE : MACHINE LEARNING

COURSE CODE : 25SC2107E

PROJECT TITLE:

Insider Threat Detection Using Elastic Net Regularized Logistic Regression

Abstract

Insider threats are a major cybersecurity concern because authorized users can misuse their access to steal sensitive information or harm organizational systems. This project proposes an **Insider Threat Detection System using Elastic Net Regularized Logistic Regression** to identify suspicious user activities based on behavioral and security-related data. The system analyzes features such as login patterns, file access, access frequency, and unusual user behavior to classify activities as normal or potentially malicious.

Elastic Net regularization combines L1 and L2 penalties, helping select relevant features and reduce overfitting when dealing with correlated data. The model will be trained and evaluated using performance measures such as accuracy, precision, recall, F1-score, and ROC-AUC. The proposed system demonstrates how machine learning can assist cybersecurity teams in identifying suspicious behavior early and reducing the risk of insider attacks.

TEAM MEMBERS:

2520030436 – G. Eesha Yuktha

2520030258 – Sai Saranya